

Raghav Ramani

GitHub | LinkedIn | Portfolio | raghavramani2004@gmail.com | +91 8003708227 | Jaipur, India

SUMMARY

Full-stack AI/ML engineer specializing in multi-agent orchestration, production RAG pipelines, and full-stack GenAI applications. Builds enterprise and GovTech platforms with end-to-end vector ingestion, LangGraph execution, FastAPI backends, Next.js UIs, and live database integrations.

WORK EXPERIENCE

Aasvaa Innovation Labs

Jan 2026–Present

Machine Learning & AI Engineer

Jaipur

- Co-developed the GovTech Planning Intelligence Platform (“Ask Manthaa OS”) full-stack features + AI chatbot layer over live Supabase data across 41 districts.
- Architected a 7-stage multi-agent Legal AI engine (FastAPI + LangGraph + Groq Llama 3.1 70B) with hybrid retrieval (Tavily real-time search + Qdrant over 70K+ Supreme Court vectors) achieving a 3.2% hallucination rate and <2.2s latency.

PockyDoc

Apr–Jun 2026

AI Product Engineer Intern

Remote

- Architected end-to-end AI system design and led project execution, aligning technical delivery with product roadmap strategy.
- Conducted Exploratory Data Analysis (EDA) on core datasets, extracting data-driven business insights to shape feature development.

Klear AI

Aug–Nov 2025

AI/ML Engineer Intern

- Built production RAG pipelines (LangChain + Qdrant + Groq Llama 3.1 70B) with embedding optimization and custom retrieval tuning.
- Developed GenAI workflows: prompt-engineering pipelines, CNN image-detection models, and evaluation benchmarks.

Accio Job

Teaching Assistant (Java)

- Co-delivered Java + OOP curriculum to 50 students; led code reviews, doubt sessions, and technical workshops.

PROJECTS

GovTech Planning Intelligence Platform

Aug 2023–Feb 2024

Live

- District-development planning platform aggregating governance data across 41 districts, 457 blocks, and 14,000+ local bodies for a state government initiative.
- Built “Ask Manthaa OS”: full-stack Hindi/English NLP assistant over 11.87L aspirations, automating district-level strategic briefs (42 PDFs) and 9-sector KPI dashboards at 95.87% uptime.
- Engineered an aspiration-driven budget allocation engine computing optimal fund distribution by criticality score, baseline deficit, and funding status per sector.

SarkariSaathi AI GovTech Platform

GitHub | Live

- AI-powered GovTech platform helping Indian citizens navigate government services across 30 states without paid middlemen.
- Built on Next.js 14 + Supabase with full Hindi/English i18n, seeding 27+ state-specific guides via SQL migration; Gemini intent detection, BetterAuth, and ArcJet rate limiting.
- Cut query-to-answer time from 15–20s to 2–3s (80–85% reduction); SEO-optimized structured metadata achieved full search-engine indexing.

Legal AI System

GitHub

- 7-stage deterministic multi-agent research engine (LangGraph + Groq Llama 3.1 70B) over 70K+ Indian Supreme Court judgments.
- Hybrid retrieval: live web research via Tavily API fused with semantic vector search via Qdrant.
- Confidence-scored outputs [30–95], self-healing JSON parsing, hallucination guard, 0.91 retrieval precision, and 3.2% hallucination rate.

ArthMitra — AI-Powered Financial Security Platform

GitHub

- LangGraph Supervisor routing a 3-agent swarm (Auditor/Shield/Mitra) across Deepseek-R1, Qwen2.5, Gemma3, and Groq Llama-3.1.
- Shield ML: TF-IDF+LR, Random Forest, and One-Class SVM for real-time SMS-scam detection, transaction-risk scoring, and credit profiling; Next.js glassmorphism UI.

SKILLS

- Languages:** Python, SQL, JavaScript (intermediate)
- Backend/DB:** FastAPI, Flask, Async Python, Supabase, PostgreSQL, REST APIs
- AI/GenAI:** LangChain, LangGraph, MCP, RAG, Multi-Agent Systems, Prompt Engineering, Groq, Gemini, Ollama
- Vector/NLP:** Qdrant, ChromaDB, FAISS, Sentence Transformers, spaCy, NLTK

EDUCATION

JECRC University	2022–2026
B.Tech CSE (AI & ML) Specialization: Artificial Intelligence & Machine Learning	CGPA: 7.2

CERTIFICATIONS & ACHIEVEMENTS

- **LangChain Academy:** 9 certifications in LangChain, LangGraph, LangSmith, agentic architectures.
- **SQL for Data Analysis:** Intellipaat (joins, window functions, subqueries, aggregations).
- **GSEA Winner:** Suraksha Nivesh AI stock-scam prediction project.